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United States Patent
Kind Code
Date of Patent
Inventor(s)

12408757
B2
September 09, 2025
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Shooting rest chair

Abstract

A shooting rest chair, components thereof, and associated methods. The shooting rest chair includes a base configured to rest on a support surface. A seat supported by the base is sized and shaped to support a shooter when the shooter is sitting in the seat. A pivot hub is supported by the base and connects the base to the seat. A weapon support arm assembly connects to the pivot hub by a first pivot connection. The weapon support arm assembly is configured to support the weapon and is pivotable with respect to the base and the seat about the first pivot connection.

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Family ID: 83601964

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Appl. No.: 18/653117

Filed: May 02, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240277149 A1	Aug. 22, 2024

Related U.S. Application Data

continuation parent-doc US 17659222 20220414 US 12004658 child-doc US 18653117
us-provisional-application US 63175361 20210415

Publication Classification

Int. Cl.: A01M31/02 (20060101); A47C3/18 (20060101); A47C4/44 (20060101); A47C4/52 (20060101); A47C7/62 (20060101)

U.S. Cl.:

CPC A47C4/52 (20130101); A01M31/02 (20130101); A47C3/18 (20130101); A47C4/44 (20130101); A47C7/62 (20130101);

Field of Classification Search

CPC: F41A (23/02); A01M (31/02); A47C (3/18); A47C (4/44); A47C (4/52); A47C (9/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
197397	12/1876	O'Neil	N/A	N/A
387411	12/1887	Gisel	N/A	N/A
399604	12/1888	Dufner et al.	N/A	N/A
499315	12/1892	Borchardt	N/A	N/A
568543	12/1895	Parks	N/A	N/A
668219	12/1900	Rock	N/A	N/A
691912	12/1901	McClean	N/A	N/A
718865	12/1902	Northcraft	N/A	N/A
778865	12/1904	Hyenga	N/A	N/A
789909	12/1904	Herold	N/A	N/A
1033624	12/1911	Schmeisser	N/A	N/A
1061577	12/1912	Whitney	N/A	N/A
1088362	12/1913	Perkins	N/A	N/A
1089307	12/1913	Benet et al.	N/A	N/A
1121945	12/1913	Smith	N/A	N/A
1145585	12/1914	Hebard	N/A	N/A
1175692	12/1915	Boicourt	N/A	N/A
1187325	12/1915	Ivey	N/A	N/A
1195777	12/1915	Burtin	N/A	N/A
1250215	12/1916	Panos	N/A	N/A
1256255	12/1917	Porter	N/A	N/A
1295688	12/1918	Butler	N/A	N/A
1367353	12/1920	Craig	N/A	N/A
1457407	12/1922	Stokes	N/A	N/A
1488647	12/1923	Quinn	N/A	N/A
1491604	12/1923	Fuller	N/A	N/A
1499748	12/1923	Papouchis	N/A	N/A
1639722	12/1926	Whitney	N/A	N/A
1693289	12/1927	Warren	N/A	N/A
1736244	12/1928	Baker	N/A	N/A
1902040	12/1932	Meyer	N/A	N/A
1907181	12/1932	Fey	N/A	N/A
1927876	12/1932	Meyer	N/A	N/A
1928871	12/1932	Swebilius	N/A	N/A
1938345	12/1932	Monchiero et al.	N/A	N/A

2066218	12/1935	Morgan	N/A	N/A
2079510	12/1936	King et al.	N/A	N/A
2090930	12/1936	Chubb	N/A	N/A
2100514	12/1936	Miller	N/A	N/A
2121982	12/1937	Pugsley	N/A	N/A
2125353	12/1937	Mattson	N/A	N/A
2216766	12/1939	Cook	N/A	N/A
2228179	12/1940	Motley	N/A	N/A
2232743	12/1940	Swenson	N/A	N/A
2297993	12/1941	Tratsch	N/A	N/A
2331372	12/1942	Buchanan	N/A	N/A
2378545	12/1944	Fraser et al.	N/A	N/A
D147305	12/1946	Sloan	N/A	N/A
2427365	12/1946	Meister	N/A	N/A
2432519	12/1946	Garand	N/A	N/A
2451266	12/1947	Whittemore	N/A	N/A
2455644	12/1947	Barnes	N/A	N/A
2476078	12/1948	Banks	N/A	N/A
2479354	12/1948	Hanson	N/A	N/A
2483089	12/1948	Ferguson	N/A	N/A
2484801	12/1948	Anderson	N/A	N/A
2508951	12/1949	Kazimier	N/A	N/A
2510380	12/1949	Clifford	N/A	N/A
2517268	12/1949	Wilson	N/A	N/A
2572585	12/1950	Barber	N/A	N/A
2582140	12/1951	Leek	N/A	N/A
2638676	12/1952	Callahan	N/A	N/A
2677207	12/1953	Stewart	N/A	N/A
2701009	12/1954	Richard	297/440.16	A47C 9/10
2701930	12/1954	Dolan	N/A	N/A
2729276	12/1955	Volney	297/34	A47C 4/52
2729975	12/1955	Hawthornet et al.	N/A	N/A
2731829	12/1955	Wigington et al.	N/A	N/A
2740530	12/1955	Ponder	N/A	N/A
2753642	12/1955	Sullivan	N/A	N/A
2774090	12/1955	Allinson	N/A	N/A
2774563	12/1955	Pribis	N/A	N/A
2795881	12/1956	Bellows	N/A	N/A
2813376	12/1956	Middlemark	N/A	N/A
2817233	12/1956	Dower et al.	N/A	N/A
2821117	12/1957	Hultgren	N/A	N/A
2847909	12/1957	Kester	N/A	N/A
2867931	12/1958	Schreiber	N/A	N/A
2874707	12/1958	Koppel	N/A	N/A
2877689	12/1958	Pribis	N/A	N/A
2894347	12/1958	Woodcock	N/A	N/A
2924881	12/1959	Gee	N/A	N/A
2924904	12/1959	Amsler	N/A	N/A
2924914	12/1959	Garwood	N/A	N/A
2973803	12/1960	Mickelson	N/A	N/A
2975540	12/1960	Lewis	N/A	N/A

2999788	12/1960	Morgan	N/A	N/A
3011283	12/1960	Lunn et al.	N/A	N/A
3012350	12/1960	Wold	N/A	N/A
3013289	12/1960	Sasena	N/A	N/A
3023527	12/1961	Leek et al.	N/A	N/A
3024653	12/1961	Broadway	N/A	N/A
3041938	12/1961	Seabrook	N/A	N/A
3055655	12/1961	Chelf	N/A	N/A
3060612	12/1961	Brown et al.	N/A	N/A
3064976	12/1961	Kuhn	N/A	N/A
3107642	12/1962	Lakin	N/A	N/A
3112567	12/1962	Flanagan	N/A	N/A
3125929	12/1963	Peasley	N/A	N/A
3128668	12/1963	Dicken	N/A	N/A
3137957	12/1963	Ingalls	N/A	N/A
3163420	12/1963	Braun	N/A	N/A
3175456	12/1964	Goodsell	N/A	N/A
3183617	12/1964	Ruger et al.	N/A	N/A
3205518	12/1964	Romaine	N/A	N/A
3206885	12/1964	Dye	N/A	N/A
3225656	12/1964	Flaherty et al.	N/A	N/A
D203680	12/1965	Allison	N/A	N/A
3240103	12/1965	Lamont	N/A	N/A
3259986	12/1965	Carr	N/A	N/A
3283425	12/1965	Boyd	N/A	N/A
3283643	12/1965	Mittelsteadt	N/A	N/A
3291317	12/1965	Bowen	N/A	N/A
3292293	12/1965	Chiasera et al.	N/A	N/A
3320848	12/1966	Ponsness	N/A	N/A
3323246	12/1966	Loffler	N/A	N/A
3327422	12/1966	Harris	N/A	N/A
3330561	12/1966	Kandel	N/A	N/A
3343411	12/1966	Lee	N/A	N/A
3349728	12/1966	Barecki et al.	N/A	N/A
3353827	12/1966	Dun, Jr.	N/A	N/A
3358504	12/1966	Freebairn	N/A	N/A
3370852	12/1967	Kandel	N/A	N/A
3406969	12/1967	Tisdell et al.	N/A	N/A
3423092	12/1968	Kandel	N/A	N/A
D215311	12/1968	Born	N/A	N/A
3473673	12/1968	Porter	N/A	N/A
3474743	12/1968	Blevins	N/A	N/A
3486752	12/1968	Colvin	N/A	N/A
3499525	12/1969	Kanter	N/A	N/A
3510951	12/1969	Dow	N/A	N/A
3513604	12/1969	Matsunaga et al.	N/A	N/A
3536160	12/1969	Brewer	N/A	N/A
3550941	12/1969	Spiro et al.	N/A	N/A
3556666	12/1970	Lichenstern	N/A	N/A
D220154	12/1970	Irelan	N/A	N/A
3572712	12/1970	Vick	N/A	N/A

3580127	12/1970	Lee	N/A	N/A
3583556	12/1970	Wagner	N/A	N/A
3584820	12/1970	Butcher, Sr.	N/A	N/A
3587193	12/1970	Lewis	N/A	N/A
3608225	12/1970	Manuel	N/A	N/A
3609902	12/1970	Casull	N/A	N/A
3646704	12/1971	Ellsworth	N/A	N/A
3648909	12/1971	Wisecarver	N/A	N/A
3680266	12/1971	Shiplov	N/A	N/A
3680354	12/1971	Phillips, Jr.	N/A	N/A
3685194	12/1971	Coon	N/A	N/A
3711955	12/1972	Holt	N/A	N/A
3711984	12/1972	Dyer et al.	N/A	N/A
3736243	12/1972	Duggan	N/A	N/A
3738101	12/1972	Simon-Vermot	N/A	N/A
3739515	12/1972	Koon, Jr.	N/A	N/A
3743088	12/1972	Henkin	N/A	N/A
3744292	12/1972	Michelson	N/A	N/A
3745875	12/1972	Kennedy et al.	N/A	N/A
3748950	12/1972	Huntington	N/A	N/A
3764219	12/1972	Collins	N/A	N/A
3769758	12/1972	McDonald	N/A	N/A
3771176	12/1972	Herman, Sr.	N/A	N/A
3804238	12/1973	Howard	N/A	N/A
3813816	12/1973	Funk	N/A	N/A
3815270	12/1973	Pachmayr	N/A	N/A
3826559	12/1973	Berliner et al.	N/A	N/A
3827172	12/1973	Howe	N/A	N/A
3842527	12/1973	Low	N/A	N/A
D233853	12/1973	Ferrara	N/A	N/A
3876078	12/1974	Gomes et al.	N/A	N/A
3877178	12/1974	Campanelli	N/A	N/A
3878939	12/1974	Wilcox	N/A	N/A
3885357	12/1974	Hoyt	N/A	N/A
3893266	12/1974	Anderson et al.	N/A	N/A
3895803	12/1974	Loe	N/A	N/A
3899175	12/1974	Loe	N/A	N/A
3899797	12/1974	Gunst	N/A	N/A
D237106	12/1974	Baljet et al.	N/A	N/A
3913746	12/1974	Burton	N/A	N/A
3914879	12/1974	Taylor, III et al.	N/A	N/A
3935657	12/1975	Wade	N/A	N/A
3947988	12/1975	Besaw	N/A	N/A
3949987	12/1975	Candor	N/A	N/A
3961436	12/1975	Hagen et al.	N/A	N/A
3964613	12/1975	Anderson, Jr.	N/A	N/A
3979849	12/1975	Haskins	N/A	N/A
4007554	12/1976	Helmstadter	N/A	N/A
4012860	12/1976	Auger	N/A	N/A
4018339	12/1976	Pritz	N/A	N/A
4021971	12/1976	McFadden	N/A	N/A

4026057	12/1976	Cady	N/A	N/A
4027781	12/1976	Covert	N/A	N/A
4042242	12/1976	Nicholls et al.	N/A	N/A
4054288	12/1976	Perrine, Sr.	N/A	N/A
4055016	12/1976	Katsenes	N/A	N/A
4072313	12/1977	Murso et al.	N/A	N/A
4076247	12/1977	Kim et al.	N/A	N/A
4090606	12/1977	Dawson	N/A	N/A
4120108	12/1977	Vickers et al.	N/A	N/A
4120276	12/1977	Curran	N/A	N/A
4122623	12/1977	Stice	N/A	N/A
4143491	12/1978	Blanc	N/A	N/A
4177608	12/1978	Balz	N/A	N/A
4188855	12/1979	Alberts	N/A	N/A
4203600	12/1979	Brown	N/A	N/A
4206573	12/1979	Hayward	N/A	N/A
4207699	12/1979	Hensley	N/A	N/A
4222305	12/1979	Lee	N/A	N/A
4223588	12/1979	Simpson	N/A	N/A
4233748	12/1979	Ford et al.	N/A	N/A
D257687	12/1979	Bechtel	N/A	N/A
4265045	12/1980	Garbini	N/A	N/A
4266748	12/1980	Dalton	N/A	N/A
4266780	12/1980	McQuary	N/A	N/A
4282671	12/1980	Wood et al.	N/A	N/A
D260650	12/1980	Alviti	N/A	N/A
4296963	12/1980	Blanchard et al.	N/A	N/A
D261794	12/1980	Bechtel	N/A	N/A
4301625	12/1980	Rampe	N/A	N/A
4312146	12/1981	Koon, Jr. et al.	N/A	N/A
4332185	12/1981	Hargrove	N/A	N/A
4333385	12/1981	Culver	N/A	N/A
4338726	12/1981	Swales	N/A	N/A
4340370	12/1981	Marshall et al.	N/A	N/A
4345398	12/1981	Pickett	N/A	N/A
4346530	12/1981	Stewart et al.	N/A	N/A
4359833	12/1981	Pachmayr et al.	N/A	N/A
4361989	12/1981	Ohno	N/A	N/A
4385464	12/1982	Casull	N/A	N/A
4385545	12/1982	Duer	N/A	N/A
4391058	12/1982	Casull	N/A	N/A
4392321	12/1982	Bosworth	N/A	N/A
4407379	12/1982	Pryor et al.	N/A	N/A
4409751	12/1982	Goda et al.	N/A	N/A
4409826	12/1982	Wenger	N/A	N/A
4426085	12/1983	Dixon	N/A	N/A
4438581	12/1983	LaValle	N/A	N/A
4438913	12/1983	Hylla	N/A	N/A
4446900	12/1983	Markovich	N/A	N/A
4449314	12/1983	Sorensen	N/A	N/A
4462598	12/1983	Chalin et al.	N/A	N/A

4477082	12/1983	McKenzie et al.	N/A	N/A
4480411	12/1983	Blaz et al.	N/A	N/A
4501071	12/1984	Manske	N/A	N/A
4506466	12/1984	Hall	N/A	N/A
4508508	12/1984	Theodore	N/A	N/A
4512101	12/1984	Waterman, Jr.	N/A	N/A
4522102	12/1984	Pickens	N/A	N/A
4526084	12/1984	David et al.	N/A	N/A
4535559	12/1984	Hall	N/A	N/A
4540182	12/1984	Clement	N/A	N/A
4542677	12/1984	Lee	N/A	N/A
4545144	12/1984	Schuster	N/A	N/A
4548392	12/1984	Rickling	N/A	N/A
4558531	12/1984	Kilby	N/A	N/A
4565403	12/1985	Brown	N/A	N/A
D283561	12/1985	Geist et al.	N/A	N/A
4601124	12/1985	Brown, Jr.	N/A	N/A
4608762	12/1985	Varner	N/A	N/A
4621563	12/1985	Poiencot	N/A	N/A
4625620	12/1985	Harris	N/A	N/A
4632008	12/1985	Homer	N/A	N/A
4644987	12/1986	Kiang et al.	N/A	N/A
4645167	12/1986	Hardwick	248/283.1	B63B 29/04
4648191	12/1986	Goff et al.	N/A	N/A
4653210	12/1986	Poff, Jr.	N/A	N/A
4671364	12/1986	Fink et al.	N/A	N/A
4674216	12/1986	Ruger et al.	N/A	N/A
4695060	12/1986	Pilgrim	N/A	N/A
4696356	12/1986	Ellion et al.	N/A	N/A
4702029	12/1986	Shaine	N/A	N/A
4715476	12/1986	France	N/A	N/A
4715499	12/1986	Franklin	N/A	N/A
4716673	12/1987	Williams et al.	N/A	N/A
4721205	12/1987	Burt et al.	N/A	N/A
4723472	12/1987	Lee	N/A	N/A
4729186	12/1987	Rieger	N/A	N/A
4732394	12/1987	Stein et al.	N/A	N/A
4736843	12/1987	Leonard	N/A	N/A
4739996	12/1987	Vedder	N/A	N/A
4751963	12/1987	Bui et al.	N/A	N/A
D297855	12/1987	Ruger et al.	N/A	N/A
4776471	12/1987	Elkins	N/A	N/A
4790079	12/1987	Meyers	N/A	N/A
4790096	12/1987	Gibson et al.	N/A	N/A
4798411	12/1988	Lin	N/A	N/A
4799324	12/1988	Nodo	N/A	N/A
4807381	12/1988	Southard	N/A	N/A
4807888	12/1988	Pidde et al.	N/A	N/A
4815593	12/1988	Brown	N/A	N/A
4819359	12/1988	Bassett	N/A	N/A
4821256	12/1988	Schmidt	N/A	N/A

4821422	12/1988	Porter	N/A	N/A
4821443	12/1988	Bianco et al.	N/A	N/A
4823673	12/1988	Downing	N/A	N/A
4824086	12/1988	Rickling, deceased et al.	N/A	N/A
4841839	12/1988	Stuart	N/A	N/A
4850151	12/1988	Ditscherlein	N/A	N/A
4854066	12/1988	Canterbury, Sr.	N/A	N/A
4862567	12/1988	Beebe	N/A	N/A
D304223	12/1988	Ruger et al.	N/A	N/A
4873777	12/1988	Southard	N/A	N/A
4877131	12/1988	Patros et al.	N/A	N/A
4890406	12/1989	French	N/A	N/A
4890847	12/1989	Cartee et al.	N/A	N/A
4896446	12/1989	Gregory	N/A	N/A
D306234	12/1989	Ferstl et al.	N/A	N/A
4903425	12/1989	Harris	N/A	N/A
4910904	12/1989	Rose	N/A	N/A
4918825	12/1989	Lesh et al.	N/A	N/A
4923402	12/1989	Marshall et al.	N/A	N/A
4924616	12/1989	Bell	N/A	N/A
4937965	12/1989	Narvaez	N/A	N/A
D310302	12/1989	Southard	N/A	N/A
4967497	12/1989	Yakscoe	N/A	N/A
4971208	12/1989	Reinfried, Jr. et al.	N/A	N/A
4972619	12/1989	Eckert	N/A	N/A
4979752	12/1989	Fosseen	N/A	N/A
D313886	12/1990	Southard	N/A	N/A
4987694	12/1990	Lombardo	N/A	N/A
4998367	12/1990	Leibowitz	N/A	N/A
4998944	12/1990	Lund	N/A	N/A
5005657	12/1990	Ellion et al.	N/A	N/A
5009021	12/1990	Nelson	N/A	N/A
5014793	12/1990	Germanton et al.	N/A	N/A
5031348	12/1990	Carey	N/A	N/A
5050330	12/1990	Pilgrim et al.	N/A	N/A
5058302	12/1990	Minneman	N/A	N/A
5060410	12/1990	Mueller	N/A	N/A
5063679	12/1990	Schwandt	N/A	N/A
5067268	12/1990	Ransom	N/A	N/A
5070636	12/1990	Mueller	N/A	N/A
5074188	12/1990	Harris	N/A	N/A
5081783	12/1991	Jarvis	N/A	N/A
5117850	12/1991	Money	N/A	N/A
5123194	12/1991	Mason	N/A	N/A
5125389	12/1991	Paff	N/A	N/A
5143340	12/1991	Wood et al.	N/A	N/A
5149900	12/1991	Buck	N/A	N/A
5173563	12/1991	Gray	N/A	N/A
5180874	12/1992	Troncoso, Jr.	N/A	N/A
5185927	12/1992	Rivers	N/A	N/A

5186468	12/1992	Davies	N/A	N/A
5188371	12/1992	Edwards	N/A	N/A
5194678	12/1992	Kramer	N/A	N/A
D335896	12/1992	Evenson	N/A	N/A
5211404	12/1992	Grant	N/A	N/A
5221806	12/1992	Chaney et al.	N/A	N/A
5222306	12/1992	Neumann	N/A	N/A
5228887	12/1992	Mayer et al.	N/A	N/A
5232227	12/1992	Bateman	N/A	N/A
5233779	12/1992	Shaw	N/A	N/A
5235764	12/1992	Perazzi	N/A	N/A
5237778	12/1992	Baer	N/A	N/A
5240258	12/1992	Bateman	N/A	N/A
5247758	12/1992	Mason	N/A	N/A
5271175	12/1992	West, III	N/A	N/A
5275890	12/1993	Wolf et al.	N/A	N/A
5287643	12/1993	Arizpe-Gilmore	N/A	N/A
5311693	12/1993	Underwood	N/A	N/A
5315781	12/1993	Beisner	N/A	N/A
5316579	12/1993	McMillan et al.	N/A	N/A
5320217	12/1993	Lenarz	N/A	N/A
5320223	12/1993	Allen	N/A	N/A
5328029	12/1993	Chow et al.	N/A	N/A
5332185	12/1993	Walker, III	N/A	N/A
5333829	12/1993	Bell et al.	N/A	N/A
5335578	12/1993	Lorden et al.	N/A	N/A
5337505	12/1993	Brown et al.	N/A	N/A
5344012	12/1993	Matthews	N/A	N/A
5347740	12/1993	Rather et al.	N/A	N/A
5351428	12/1993	Graham	N/A	N/A
5354247	12/1993	Wilkinson	N/A	N/A
5358254	12/1993	Yeh et al.	N/A	N/A
5361505	12/1993	Faughn	N/A	N/A
5367232	12/1993	Netherton et al.	N/A	N/A
5370240	12/1993	Hand	N/A	N/A
5375337	12/1993	Butler	N/A	N/A
5375377	12/1993	Kenton	N/A	N/A
5392553	12/1994	Carey	N/A	N/A
5394983	12/1994	Latulippe et al.	N/A	N/A
5402595	12/1994	Tamillos	N/A	N/A
5406733	12/1994	Tarlton et al.	N/A	N/A
5410833	12/1994	Paterson	N/A	N/A
5414949	12/1994	Peebles	N/A	N/A
D359392	12/1994	Bellington	N/A	N/A
5421115	12/1994	McKay	N/A	N/A
5433010	12/1994	Bell	N/A	N/A
5433451	12/1994	DeVries	N/A	N/A
5435223	12/1994	Blodgett et al.	N/A	N/A
5442860	12/1994	Palmer	N/A	N/A
D362116	12/1994	Bellington et al.	N/A	N/A
5446987	12/1994	Lee et al.	N/A	N/A

D364080	12/1994	Weyrauch	N/A	N/A
5481817	12/1995	Parker	N/A	N/A
5482241	12/1995	Oglesby	N/A	N/A
5486135	12/1995	Arpaio	N/A	N/A
5490302	12/1995	Dion	N/A	N/A
5491921	12/1995	Allen	N/A	N/A
5497557	12/1995	Martinsson et al.	N/A	N/A
5497575	12/1995	Fried et al.	N/A	N/A
5501467	12/1995	Kandel	N/A	N/A
D369904	12/1995	Taylor	N/A	N/A
5525314	12/1995	Hurson	N/A	N/A
5540329	12/1995	Vogeley	N/A	N/A
5545855	12/1995	Stanfield et al.	N/A	N/A
5562208	12/1995	Hasler et al.	N/A	N/A
D375538	12/1995	Minneman	N/A	N/A
5570513	12/1995	Peterson	N/A	N/A
5580063	12/1995	Edwards	N/A	N/A
5588242	12/1995	Hughes	N/A	N/A
D377823	12/1996	Rainwater	N/A	N/A
5600913	12/1996	Minneman	N/A	N/A
5617666	12/1996	Scott	N/A	N/A
5622344	12/1996	Gracie	N/A	N/A
5628135	12/1996	Cady	N/A	N/A
D380116	12/1996	Minneman	N/A	N/A
5640944	12/1996	Minneman	N/A	N/A
5644862	12/1996	Folmer	N/A	N/A
5649465	12/1996	Beebe	N/A	N/A
5651207	12/1996	Knight	N/A	N/A
5653625	12/1996	Pierce et al.	N/A	N/A
5661919	12/1996	Pryor	N/A	N/A
5662516	12/1996	You	N/A	N/A
5666757	12/1996	Helmstadter	N/A	N/A
5688024	12/1996	Arizpe-Gilmore	297/440.1	A01M 31/02
D387123	12/1996	Hughes et al.	N/A	N/A
5697180	12/1996	Morizio	N/A	N/A
5703317	12/1996	Levilly et al.	N/A	N/A
5704482	12/1997	Apps et al.	N/A	N/A
5711102	12/1997	Plaster et al.	N/A	N/A
5711103	12/1997	Keng	N/A	N/A
5715624	12/1997	Hilbelink et al.	N/A	N/A
5715625	12/1997	West, III	N/A	N/A
D391616	12/1997	Plybon	N/A	N/A
5723183	12/1997	Williams et al.	N/A	N/A
5723806	12/1997	Odom	N/A	N/A
5725096	12/1997	Winnard	N/A	N/A
5737865	12/1997	Brandl et al.	N/A	N/A
5740625	12/1997	Jenkins	N/A	N/A
5743395	12/1997	Backer	N/A	N/A
5758447	12/1997	Venetz	N/A	N/A
5758933	12/1997	Clendening	N/A	N/A
5761954	12/1997	Dvorak	N/A	N/A

5778589	12/1997	Teague	N/A	N/A
5779527	12/1997	Maebashi	N/A	N/A
5791499	12/1997	Zebbedies	N/A	N/A
5811720	12/1997	Quinnell et al.	N/A	N/A
5815974	12/1997	Keng	N/A	N/A
5833308	12/1997	Strong, III et al.	N/A	N/A
D403176	12/1997	Harper	N/A	N/A
5845774	12/1997	Hausknecht	N/A	N/A
5857279	12/1998	de Oliveira Masina et al.	N/A	N/A
5875580	12/1998	Hill et al.	N/A	N/A
5878504	12/1998	Harms	N/A	N/A
5884966	12/1998	Hill et al.	N/A	N/A
5899329	12/1998	Hu et al.	N/A	N/A
5907919	12/1998	Keeney	N/A	N/A
5913131	12/1998	Hossain et al.	N/A	N/A
5913422	12/1998	Cote et al.	N/A	N/A
5913667	12/1998	Smilee	N/A	N/A
5913668	12/1998	Messer	N/A	N/A
5924694	12/1998	Kent	N/A	N/A
5930932	12/1998	Peterson	N/A	N/A
5933997	12/1998	Barrett	N/A	N/A
5933999	12/1998	McClure et al.	N/A	N/A
5937561	12/1998	Abernethy	N/A	N/A
5959613	12/1998	Rosenbreg et al.	N/A	N/A
5970642	12/1998	Martin	N/A	N/A
5974719	12/1998	Simonek	N/A	N/A
6019375	12/1999	West, Jr.	N/A	N/A
6021891	12/1999	Anderson	N/A	N/A
6032796	12/1999	Hopper et al.	N/A	N/A
6042080	12/1999	Shepherd et al.	N/A	N/A
6044747	12/1999	Felts	N/A	N/A
6058641	12/1999	Vecqueray	N/A	N/A
6073381	12/1999	Farrar et al.	N/A	N/A
6086375	12/1999	Legros	N/A	N/A
6092662	12/1999	Frederick, Jr. et al.	N/A	N/A
6110020	12/1999	Rolfi	N/A	N/A
6121556	12/1999	Cole	N/A	N/A
6237462	12/2000	hawkes et al.	N/A	N/A
6253482	12/2000	Peterson	N/A	N/A
6254100	12/2000	Rinehart	N/A	N/A
6260463	12/2000	Brand et al.	N/A	N/A
6269578	12/2000	Callegari	N/A	N/A
6283428	12/2000	Maples et al.	N/A	N/A
6289622	12/2000	Desch, Jr. et al.	N/A	N/A
6293041	12/2000	Weaver	N/A	N/A
6294759	12/2000	Dunn, Jr.	N/A	N/A
6305117	12/2000	Hales, Sr.	N/A	N/A
6309476	12/2000	Ravenscroft et al.	N/A	N/A
6338218	12/2001	Hegler	N/A	N/A
6347831	12/2001	Nye et al.	N/A	N/A

6390294	12/2001	Fiore et al.	N/A	N/A
6397720	12/2001	Fox et al.	N/A	N/A
6439515	12/2001	Powers	N/A	N/A
6439530	12/2001	Shoenfish et al.	N/A	N/A
6517133	12/2002	Seegmiller et al.	N/A	N/A
D471248	12/2002	Jacobs	N/A	N/A
6526687	12/2002	Looney	N/A	N/A
D473376	12/2002	Abate	N/A	N/A
6539660	12/2002	Yeargin	N/A	N/A
6546662	12/2002	Chong	N/A	N/A
6557855	12/2002	Wu	N/A	N/A
6574899	12/2002	Mostello	N/A	N/A
6575469	12/2002	Lowe	N/A	N/A
6637708	12/2002	Peterson	182/187	A01M 31/02
6643973	12/2002	Smith	N/A	N/A
6663298	12/2002	Haney	N/A	N/A
6688031	12/2003	Steele	N/A	N/A
6698830	12/2003	Gaines	N/A	N/A
6733375	12/2003	Hoffman	N/A	N/A
6736400	12/2003	Cesternino	N/A	N/A
6813855	12/2003	Pinkley	N/A	N/A
6814654	12/2003	Rolfi	N/A	N/A
6854975	12/2004	Ranzinger	N/A	N/A
6860054	12/2004	Mosher	N/A	N/A
6860055	12/2004	Walrath	N/A	N/A
6862833	12/2004	Gutner	N/A	N/A
6871440	12/2004	Highfill et al.	N/A	N/A
6877266	12/2004	Brownlee	N/A	N/A
6883263	12/2004	Carrow	N/A	N/A
6895709	12/2004	Krien et al.	N/A	N/A
6931777	12/2004	Krien	N/A	N/A
6935064	12/2004	Thompson	248/370	F41A 23/34
6953114	12/2004	Wang et al.	N/A	N/A
D513055	12/2004	Lahti	N/A	N/A
6978569	12/2004	Williamson, IV et al.	N/A	N/A
D519183	12/2005	Minneman	N/A	N/A
7032494	12/2005	Wygant	N/A	N/A
7032965	12/2005	Howell	297/344.21	A01M 31/02
D521100	12/2005	Morrow	N/A	N/A
7043862	12/2005	Franks	N/A	N/A
D522263	12/2005	Hart	N/A	N/A
7055279	12/2005	Flores	N/A	N/A
7059670	12/2005	Mills et al.	N/A	N/A
7062979	12/2005	Day et al.	N/A	N/A
D524541	12/2005	Cauley	N/A	N/A
7086192	12/2005	Deros	N/A	N/A
7104398	12/2005	Wisecarver	N/A	N/A
7134663	12/2005	Lowe et al.	N/A	N/A
7143986	12/2005	Austin et al.	N/A	N/A
7152355	12/2005	Fitzpatrick et al.	N/A	N/A
7152358	12/2005	LeAnna	297/170	F41A 23/02

7159711	12/2006	Gardner	N/A	N/A
7165750	12/2006	McCuskey et al.	N/A	N/A
7188445	12/2006	Lehman	N/A	N/A
D540904	12/2006	Werner	N/A	N/A
7200966	12/2006	Gooder	N/A	N/A
7201376	12/2006	Kuosa	N/A	N/A
7207567	12/2006	Brown	N/A	N/A
D543604	12/2006	Minneman	N/A	N/A
7213494	12/2006	James	N/A	N/A
7216404	12/2006	Doyle	N/A	N/A
7222451	12/2006	Keng et al.	N/A	N/A
7225050	12/2006	Sutula, Jr.	N/A	N/A
7246704	12/2006	Brunson et al.	N/A	N/A
7258345	12/2006	Anderson, Jr.	N/A	N/A
D553219	12/2006	Potterfield	N/A	N/A
7281346	12/2006	Cook et al.	N/A	N/A
7281347	12/2006	Carpenter	N/A	N/A
7314248	12/2007	Mabon et al.	N/A	N/A
D567895	12/2007	Cauley	N/A	N/A
7356960	12/2007	Knitt	N/A	N/A
7356961	12/2007	Williams	N/A	N/A
7357250	12/2007	Hagemann	N/A	N/A
7363740	12/2007	Kincel	N/A	N/A
7367451	12/2007	Pendergraph et al.	N/A	N/A
7401431	12/2007	Pierce et al.	N/A	N/A
7410053	12/2007	Bowen et al.	N/A	N/A
D576245	12/2007	Potterfield et al.	N/A	N/A
7421815	12/2007	Moody et al.	N/A	N/A
7426800	12/2007	Pierce et al.	N/A	N/A
7431247	12/2007	Bobro	N/A	N/A
7438356	12/2007	Howman et al.	N/A	N/A
7481015	12/2008	Mays	N/A	N/A
7536819	12/2008	Popikow	N/A	N/A
7536820	12/2008	Wade et al.	N/A	N/A
7549247	12/2008	Reese	N/A	N/A
7584690	12/2008	Cauley	N/A	N/A
D605246	12/2008	Hobbs	N/A	N/A
7631455	12/2008	Keng et al.	N/A	N/A
7631877	12/2008	Zara	N/A	N/A
7644981	12/2009	Hensley	N/A	N/A
7654498	12/2009	Beltz	N/A	N/A
7658140	12/2009	Lombardi	N/A	N/A
7665241	12/2009	Oz	N/A	N/A
7676977	12/2009	Cahill et al.	N/A	N/A
7681886	12/2009	Morrow et al.	N/A	N/A
7694973	12/2009	Hofmeister	N/A	N/A
7713180	12/2009	Wickens et al.	N/A	N/A
7726478	12/2009	Potterfield et al.	N/A	N/A
7730824	12/2009	Black	N/A	N/A
7743544	12/2009	Laney et al.	N/A	N/A
7753441	12/2009	Gray, Jr.	N/A	N/A

7774972	12/2009	Potterfield et al.	N/A	N/A
7779572	12/2009	Potterfield et al.	N/A	N/A
7823317	12/2009	Potterfield et al.	N/A	N/A
7845267	12/2009	Potterfield et al.	N/A	N/A
7866081	12/2010	Seuk	N/A	N/A
7883396	12/2010	Potterfield et al.	N/A	N/A
7930852	12/2010	Hubbard, Jr.	N/A	N/A
7954272	12/2010	Potterfield et al.	N/A	N/A
7997021	12/2010	Cauley	N/A	N/A
8011129	12/2010	Cauley et al.	N/A	N/A
8104212	12/2011	Potterfield et al.	N/A	N/A
8123180	12/2011	Shipman et al.	N/A	N/A
8262157	12/2011	Novak	N/A	N/A
8296988	12/2011	Yale et al.	N/A	N/A
8336708	12/2011	Potterfield et al.	N/A	N/A
8371057	12/2012	Coffield et al.	N/A	N/A
8444056	12/2012	Gamez et al.	N/A	N/A
8496212	12/2012	Keng et al.	N/A	N/A
8556036	12/2012	Meredith	182/187	A01M 31/02
8621773	12/2013	Morrow et al.	N/A	N/A
D709295	12/2013	Winebrenner et al.	N/A	N/A
8979184	12/2014	Stafford et al.	N/A	N/A
D727046	12/2014	Alexander	N/A	N/A
D727047	12/2014	Alexander	N/A	N/A
9151561	12/2014	Morrow et al.	N/A	N/A
9341427	12/2015	Bricko et al.	N/A	N/A
9474376	12/2015	Drew	N/A	N/A
9618291	12/2016	Henderson	N/A	N/A
9702653	12/2016	Cauley, Jr. et al.	N/A	N/A
9743774	12/2016	Hauser	N/A	N/A
9770109	12/2016	Mailloux	N/A	N/A
10028587	12/2017	Kersey	N/A	A47C 4/52
10194751	12/2018	Frankel	N/A	A47C 9/105
10264893	12/2018	Keilhauer	N/A	A47C 7/006
10299599	12/2018	Browning et al.	N/A	N/A
D850809	12/2018	Xiao	N/A	N/A
10328322	12/2018	King	N/A	N/A
10514225	12/2018	Cauley, Jr. et al.	N/A	N/A
10578389	12/2019	Kinney et al.	N/A	N/A
10738937	12/2019	Decker	N/A	N/A
10874216	12/2019	Grace	N/A	N/A
11166563	12/2020	Chen	N/A	A47B 39/00
2002/0195752	12/2001	Yang	N/A	N/A
2003/0234205	12/2002	McGuyer et al.	N/A	N/A
2004/0112777	12/2003	Huang	N/A	N/A
2004/0134113	12/2003	Deros et al.	N/A	N/A
2005/0000141	12/2004	Cauley et al.	N/A	N/A
2005/0115137	12/2004	Minneman	N/A	N/A
2006/0038434	12/2005	Howell et al.	N/A	N/A
2006/0065560	12/2005	Dickenson et al.	N/A	N/A
2006/0175213	12/2005	Hurt et al.	N/A	N/A

2006/0230664	12/2005	Eddins	N/A	N/A
2006/0254111	12/2005	Giauque et al.	N/A	N/A
2006/0277811	12/2005	Peterson	N/A	N/A
2007/0051028	12/2006	Stordal	N/A	N/A
2007/0068379	12/2006	Sween et al.	N/A	N/A
2007/0068835	12/2006	Buie, III	N/A	N/A
2007/0074439	12/2006	Cauley et al.	N/A	N/A
2007/0094911	12/2006	Rush et al.	N/A	N/A
2007/0113460	12/2006	Potterfield et al.	N/A	N/A
2007/0256346	12/2006	Potterfield et al.	N/A	N/A
2007/0295197	12/2006	Potterfield	N/A	N/A
2008/0018142	12/2007	Yul	N/A	N/A
2008/0023379	12/2007	Potterfield et al.	N/A	N/A
2008/0054570	12/2007	Potterfield et al.	N/A	N/A
2008/0061509	12/2007	Potterfield	N/A	N/A
2008/0128002	12/2007	Jeffs	N/A	N/A
2008/0156671	12/2007	Jansson	N/A	N/A
2008/0174071	12/2007	Potterfield et al.	N/A	N/A
2008/0178641	12/2007	Himmen	N/A	N/A
2008/0231091	12/2007	Goranson	N/A	N/A
2008/0238158	12/2007	Jiu et al.	N/A	N/A
2008/0263928	12/2007	Potterfield	N/A	N/A
2009/0025267	12/2008	Reinert et al.	N/A	N/A
2009/0049731	12/2008	Seuk	N/A	N/A
2009/0113779	12/2008	Shipman et al.	N/A	N/A
2009/0126250	12/2008	Keng	N/A	N/A
2009/0140557	12/2008	Chen	N/A	N/A
2009/0174232	12/2008	Hoffman	N/A	N/A
2009/0188146	12/2008	Werner	N/A	N/A
2010/0102178	12/2009	Smith et al.	N/A	N/A
2010/0126055	12/2009	Potterfield	N/A	N/A
2010/0138032	12/2009	Potterfield	N/A	N/A
2010/0223832	12/2009	Lombardi	N/A	N/A
2010/0236125	12/2009	Morrow et al.	N/A	N/A
2010/0270201	12/2009	Cauley et al.	N/A	N/A
2011/0024985	12/2010	Potterfield et al.	N/A	N/A
2011/0036214	12/2010	Potterfield	N/A	N/A
2011/0094140	12/2010	Letson	N/A	N/A
2011/0187164	12/2010	Corcorran et al.	N/A	N/A
2011/0192069	12/2010	Potterfield et al.	N/A	N/A
2012/0174460	12/2011	Hicks	N/A	N/A
2012/0175844	12/2011	Potterfield	N/A	N/A
2012/0186125	12/2011	Werner	N/A	N/A
2012/0267923	12/2011	Bouche	N/A	N/A
2012/0274107	12/2011	Chesness	N/A	N/A
2013/0049410	12/2012	Stafford et al.	N/A	N/A
2013/0058192	12/2012	Gateman et al.	N/A	N/A
2013/0061508	12/2012	Nelson et al.	N/A	N/A
2014/0001801	12/2013	Hutchinson	N/A	N/A
2014/0132037	12/2013	Niemela	N/A	N/A
2015/0096216	12/2014	Hughes	N/A	N/A

2015/0354913	12/2014	Morrow et al.	N/A	N/A
2016/0003571	12/2015	Kleinfelder	N/A	N/A
2016/0021869	12/2015	George	N/A	N/A
2016/0193518	12/2015	Baxter et al.	N/A	N/A
2017/0102203	12/2016	Cauley, Jr. et al.	N/A	N/A
2017/0208952	12/2016	Simon et al.	N/A	N/A
2017/0251803	12/2016	Rehkemper et al.	N/A	N/A
2017/0273451	12/2016	Jeanphilippe	N/A	N/A
2017/0340100	12/2016	Tsai	N/A	N/A
2018/0172385	12/2017	Hale	N/A	N/A
2018/0184788	12/2017	Binding	N/A	N/A
2019/0056060	12/2018	Decker	N/A	N/A
2019/0082846	12/2018	Grace	N/A	N/A
2019/0231075	12/2018	Browning et al.	N/A	N/A
2019/0285376	12/2018	Bales	N/A	N/A
2020/0033089	12/2019	Morrow et al.	N/A	N/A
2022/0265048	12/2021	Fankel et al.	N/A	N/A
2022/0386779	12/2021	Garcia	N/A	N/A
2023/0000238	12/2022	Sun	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107912904	12/2017	CN	N/A
838872	12/1951	DE	N/A
0624455	12/1993	EP	N/A
475080	12/1936	GB	N/A
101493888	12/2014	KR	N/A
WO-9519537	12/1994	WO	A01M 31/02
2016195400	12/2015	WO	N/A
2021257958	12/2020	WO	N/A
2022256552	12/2021	WO	N/A

OTHER PUBLICATIONS

Joe's, "Shooter's Ridge Steady Point Shooting Rest," <http://www.joessports.com/product/index.jsp?productld=3155005&cp=726872&parentPag...>, Item No. 3155005, 1 pg. [Internet accessed Jul. 17, 2008]. cited by applicant

Cabela's, "Sharp Shooter Rifle Rest," <http://www.cabelas.com/cabelas/en/templates/links/link.jsp?id=0005816222738a&type=product&cmCat=>, © 1996-2008, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Lahti Company, Evaluator Brochure, <http://www.lathicompany.com/Forms/EvaluatorBrochure2.jpg>, 2 pgs., accessed Jan. 16, 2006. cited by applicant

MacksPW.com, "Desert Mountain Bench Master Rifle Rest," <http://www.macksqw.com/Item-i-DESBM1>, © 2004-2008, 1 pg. [Internet accessed Jul. 22, 2008]. cited by applicant

Midway USA, "Shooters Ridge Steady Point Rifle Shooting Rest," <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=826745&t=11082005>, 2005, 5 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Sweeney, P "Gunsmithing: Measure Headspace, Peterson's Rifleshooter," http://www.rifleshootermag.com/gunsmithing/headspace_0612/, 4 pgs. [Internet Accessed Dec. 11, 2004]. cited by applicant

MidwayUSA, "ADG Rifle Shooting Rest," <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=992071&t=11082005>, 2005, 3 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

MidwayUSA, "Caldwell Full Length Fire Control Shooting Rest,"

<http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=683866&t=11082005>, 2005, 3 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 MidwayUSA, “Caldwell Lead Sled DFT Rifle Shooting Rest,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=149023&t=11082005>, 2005, 6 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 MidwayUSA, “Caldwell Lead Sled Rifle Shooting Rest,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=152664&t=11082005>, 2005, 8 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 MidwayUSA, “Caldwell Steady Rest NXT Rifle Shooting Rest,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=838651&t=11082005>, 2005, 4 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 MidwayUSA, “Caldwell Zero-Max Rifle Shooting Rest,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=726222&t=11082005>, 2005, 3 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 MidwayUSA, “CTK Precision P3 Ultimate Shooting Rest,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=114699&t=11082005>, 2005, 2 pgs. Internet accessed on 8/6/2008. cited by applicant
 MidwayUSA, “Hyskore® dangerous Game Rifle Shooting Rest,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=729197&t=11082005>, 2005, 3 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 MidwayUSA, “Hyskore® Precision Gas Dampened Recoil Reducing Rifle Shooting Rest,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=838848&t=11082005>, 2005, 3 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 MidwayUSA, “Hyskore® Swivel Varmint Rifle Shooting Rest,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=587606&t=11082005>, 2005, 3 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 MidwayUSA, “Shooters Ridge Steady Point Rifle Shooting Rest with Vise,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=341095&t=11082005>, D 2005, 4 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 MidwayUSA, “Shooting Supplies—Shop Everything for Your Firearm at MidwayUSA,” <http://www.midwayusa.com/browse/BrowseProducts.aspx?categoryStrin> . . . , 15 pgs. [Internet accessed on Jul. 21, 2008]. cited by applicant
 MidwayUSA, “Stoney Point Bench Anchor Rifle Shooting Rest,” <http://www.midwayusa.com/eproductpage.exe/showproduct?saleitemid=347174&t=11082005>, 2005, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant
 Milek, B., “Handloading for Hunting New Products from RCBS, Lee, Accurate Arms,” *Peterson's Hunting*, Mar. 1985, p. 21. 1 pg. cited by applicant
 Sinclair International, *Sinclair Shooting Rests, Products for the Precision Shooter*, 2002, Issue No. 2002-B, pp. 76-78. cited by applicant
 Shooters Ridge, “Shooting Rest with Gun Vise,” <http://www.shootersridge.com>, 1 pg. [Internet accessed Jul. 17, 2008]. cited by applicant
 Shooters Ridge, “Deluxe Rifle Rest,” <http://www.shootersridge.com>, 1 pg. [Internet accessed Jul. 21, 2008]. cited by applicant
 Protektor Model, “The Original Leather Rifle and Pistol Rest,” <http://www.protektormodel.com/>, 12 pgs. [Internet accessed on Feb. 14, 2006]. cited by applicant
 Precision Shooting, Inc., *Bald Eagle Front Rest, The Accurate Rifle*, vol. 6, Issue No. 4, May 2003, p. 47. 1 pg. cited by applicant
 “Cleaning Cradles: Sinclair Cleaning Cradles” p. 21. 1 pg. The date on which the Sinclair Folding Cleaning Cradle was first on sale is not known, but is believed to be circa 2004. cited by applicant
 “Decker Rifle Vise”, 1 pg. The date on which the Decker Rifle Vise was first on sale is not known, but is believed to be circa 2004. cited by applicant

"The Grabber and Hustler '76," MEC—Mayville Engineering Company, Inc., 2 pgs., undated. cited by applicant

1Shop2.com. "Hoppe's Gunsmith's Fully Adjustable Bench Vise", 3 pgs. The date on which The Hoppe's Gunsmith's Fully Adjustable Bench Vise was first on sale is not known, but is believed to be circa 2004. cited by applicant

AcuSport Outdoor Sporting Products, 3 pgs. cited by applicant

Cabela's: World's Foremost Outfitter. "HySkore Sighting System and Cleaning Vise". 1 pg. The date on which the HySkore Sighting System and Cleaning Vise was first on sale is not known, but is believed to be circa Jan. 2005. However, a prototype of this product may have been shown to buyers at Cabela's circa Aug 2004. cited by applicant

CV-500, 3 pgs. [product photos]. cited by applicant

Ellett Brothers, Rests & Gun Vises, 3 pgs. cited by applicant

Four photos of the Lohman Sight Vise. 4 pgs. The date on which the Lohman Sight Vise was first on sale is not known, but is believed to be circa 2004. cited by applicant

Lahti Company Brochure, "Rifle Evaluator: No Pain, No Fear, No Flinching, No Body Movement," www.lahticompany.com, 2 pgs., Undated. cited by applicant

Midway USA. "Tipton Range Box with Ultimate Rifle, Handgun Cleaning Kit (No Solvents)". <URL: <http://www.midwayusa.com/rewriteaproductU135086>>. 2 pgs. The date on which the Tipton Range Box was first on sale is not known, but is believed to be circa 2004. cited by applicant

Millett, "BenchMaster Shooting Rests," 1 pg. Undated. cited by applicant

MTM Case-Gard. "Gun Maintenance Centers." 2 pgs. The date on which the MTM Gun Maintenance Center was first on sale is not known, but is believed to be circa 2004. cited by applicant

MTM Case-Gard. "Rifle rest and pistol shooting rest". <URL: <http://www.mtmcasegard.com/products/shooting/shoo.html>>. 3 pgs. The date on which the MTM Site-In-Clean was first on sale is not known, but is believed to be circa 2004. cited by applicant

MTM Shoulder-Gard Rifle Rest, MTM Case-Gard, p. 2 "Rests", 1 pg. cited by applicant

Tenex Precision Co. "Recoil A-Rest-R" Product Pictures, 4 sheets, Riverside CA. cited by applicant

The Sportsman's Guide. "Plano Shooters Case!" <URL: <http://www.sportsmansguide.com/cb/cb.asp?a=148225>>. 3 pgs. The date on which the Plano Shooters Case was first on sale is not known, but is believed to be circa 2004. cited by applicant

Caldwell Shooting Supplies Manual, Deadshot Fieldpod, Usage and Maintenance Instructions, Product #488000—Instruction #1015026, 8 pages, published prior to Jan. 16, 2017. cited by applicant

"American Rifleman: What to do about recoil," LookSmart, http://www.findarticles.com/p/articles/mi_qa3623/is_199907/ai_n886159/print, pp. 1-4, accessed Jan. 4, 2006. cited by applicant

Hyskore: Professional Shooting Accessories, "Hydraulic Trigger Release," www.hyskore.com, 7 pgs. [Internet accessed Feb. 22, 2006]. cited by applicant

Hyskore: Professional Shooting Accessories, "Dangerous Game Machine Rest," www.hyskore.com, 10 pgs. [Internet accessed Feb. 22, 2006]. cited by applicant

Hyskore, "Rest—Dangerous Game Machine Rest," Hyskore Rest, Professional firearm rests, <http://www.hyskore.com/rests.htm>, 2 pgs. [Internet accessed Jul. 21, 2008]. cited by applicant

"Uncle Bud's Bull Bags," <http://www.unclebudscss.com/pages/Bulls%20bags.html>, 2 pgs. [Internet accessed on Feb. 14, 2006]. cited by applicant

"Uncle Bud's Udder Bag," <http://www.unclebudscss.com/pages/Udder%20Bags.html>, 2 pgs. [Internet accessed on Feb. 14, 2006]. cited by applicant

"Gun Rest-Shooting Rest-Rifle Rests," <http://www.exploreproducts.com/gunrests-shootingrests.htm>, 6 pgs. [Internet accessed Jul. 18, 2008]. cited by applicant

Grafix® Plastics, http://www.grafixplastics.com/plastic_film_g.asp?gclid=CK-5-7gnY4CFRVNhQodjFhfSQ, 29 pgs. [Internet accessed on Aug. 30, 2007]. cited by applicant

Amazon.com, "Ctk® P3 Ultimate Shooting Rest," Sports & Outdoors, <http://www.amazon.com/CTK%C2%AE-P3-Uitimate-Shooting-Rest/dp/>. . . , 1 pg. [Internet accessed on

Jul. 22, 2008]. cited by applicant

Amazon.com, "SHTRS RDG Steady PNT Rifle Rest DLX, Grips/Pads/Stocks, Gun Accessories, Hunting & Shooting Accessories, Hunting Gear, Fishing & Hunting," [http://www.amazon.com/STEADY-Accessories-Hunting-Shooting-Fishin . . .](http://www.amazon.com/STEADY-Accessories-Hunting-Shooting-Fishin...), 1 pg. [Internet accessed on Jul. 22, 2008]. cited by applicant

Amazon.com, "Stoney Point Adjustable Shooting Rest w/Bag," Sports & Outdoors, [http://www.amazon.com/Stoney-Point-Adjustable-Shooting-Rest/dp/BO ..](http://www.amazon.com/Stoney-Point-Adjustable-Shooting-Rest/dp/BO..), 1 pg. [Internet accessed on Jul. 22, 2008]. cited by applicant

Basspro.com, "Bass Pro Shops Outdoors Online: Offering the best in Fishing, Hunting and Outdoor Products," http://www.basspro.com/webapp/wcs/stores/servlet/Product_10151-1_10001_95064_SearchResults, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Battenfeld Technologies, Inc., "Gun Vise," Tipton Gun Cleaning Supplies, Battenfeld Technologies, Inc. 2004 Catalog, p. 32, Product No. 782-731, 2 pgs. cited by applicant

Battenfeld Technologies, Inc., "Steady Rest Portable Shooting Rest," file://C:\DOCUME-1\DUTC\LOCALS-1\Temp\PQ28V28J.htm, 1 pg., accessed Jan. 25, 2006. cited by applicant

Big Boy Gun Toys, "Shooting Rest," <http://www.bigboyguntoys.com/shootingrest.htm>, 1 pg. [Internet accessed on Jul. 18, 2008]. cited by applicant

Boyt Harness Company, Product Catalog, <http://www.boytharness.com/catalog/index.php?cPath=22>, 2 pgs. [Internet accessed on Jul. 21, 2008]. cited by applicant

Brownells, Inc., Catalog No. 41, 1988-1989, 3 pgs. cited by applicant

Brownells, Inc., Catalog No. 57, 2004-2005, 2 pgs. cited by applicant

Brownells, Inc., Catalog No. 47, 1994-1995, 2 pgs. cited by applicant

Brownells, Inc., Sight Base Cutters, Faxed Dec. 17, 2003, 1 pg. cited by applicant

Cabela's Hunting Fishing and Outdoor Gear Master Catalog, Fall2002, Edition II, Minimizer Rifle Rest, Item No. SC-22-4332 and SC-22-4333, p. 492. cited by applicant

Edgewood Shooting Bags Catalog, <http://www.edgebag.com/catalog.php>, 7 pgs. [Internet accessed on Feb. 14, 2006]. cited by applicant

Cabela's. "Master Catalog Fall 2003: Late-Season Edition". Cover page and p. 416. 2 pgs. cited by applicant

Cabela's, "BenchBuddy® Gun Rest," <http://www.cabelas.com/cabelas/en/templates/links/link.jsp?id=0005819221954a&type=product&cmCat=>, © 1996-2008, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Cabela's, "Elite Rifle Rest," <http://www.cabelas.com/cabelas/en/templates/links/link.jsp?id=0005817227855a&type=product&cmCat=>, © 1996-2008, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Cabela's, "Hyskore® Dangerous Game™ Machine Rest," <http://www.cabelas.com/cabelas/en/templates/links/link.jsp?id=0044091228566a&type=product&cmCat=>, © 1996-2008, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Cabela's, "Hyskore® Ultimate Sighting Rest," <http://www.cabelas.com/cabelas/en/templates/links/link.jsp?id=0024152226083a&type=product&cmCat=>, 1996-2008, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Cabela's, "Nitro Shoulder Shield Rest," <http://www.cabelas.com/cabelas/en/templates/links/link.jsp?id=0040862228231a&type=product&cmCat=>, © 1996-2008, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Cabela's, "Premier Rifle Rest," <http://www.cabelas.com/cabelas/en/templates/links/link.jsp?id=0020904227856a&type=product&cmCat=> .. , © 1996-2008, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Cabela's, "Secure Bench Rest," [http://www.cabelas.com/cabelas/en/templates/links/link.jsp;jsessionid=4FOLPOOW2HMRLLAOBBISCOF ..](http://www.cabelas.com/cabelas/en/templates/links/link.jsp;jsessionid=4FOLPOOW2HMRLLAOBBISCOF..), © 1996-2008, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Cabela's, "Sharp Shooter Auto Magnum Rifle Rest," <http://www.cabelas.com/cabelas/en/templates/links/link.jsp?id=0054107229088a&type=product&cmCat=>, © 1996-2008, 2 pgs. [Internet accessed on Aug. 6, 2008]. cited by applicant

Cabela's, "Shooting Benches & Portable Rifle Shooting Bench Rest," <http://www.cabelas.com/ssubcat-1/cat20793.shtml>, 3 pgs. [Internet accessed Jul. 18, 2008]. cited by applicant

Cabela's, "Sure Shot Shooting Vise/Rest," <http://www.cabelas.com/cabelas/en/templates/product/standard-item.jsp?id=00348272277> . . . , © 1996-2008, 2 pgs. [Internet accessed on Jul. 15, 2008]. cited by applicant

Caldwell Shooting Supplies, 2006 Catalog, pp. 18, 5, 12, 14 and 15. 5 pgs. cited by applicant

Californiavarmintcallers.com—Forum, http://californiavarmintcallers.com/community/modules/newbb/viewtopic.php?topic_id=10&forum=9&PHPSESSID=074ed8c7 . . . pages 1-4 accessed Jan. 16, 2006. cited by applicant

Canadian Camo, "Gun Rest," https://media5.magma.ca/www.canadiancamo.com/catalog/product_info.php?products_id= . . . , 2 pgs. [Internet accessed on Feb. 13, 2006]. cited by applicant

Champion Traps & Target, 2005 Product Catalog, 12 pgs. cited by applicant

CTK Precision, "P3 Ultimate Shooting Rest," <http://www.cktprecision.com/index.asp?PageAction=VIEWPROD&ProdOID=2>, 3 p. [Internet accessed on Jul. 18, 2008]. cited by applicant

CTK Precision, All Products, <http://www.cktprecision.com/index.asp?PageAction=VIEWCATS&Cate..> , 3 pgs. [Internet accessed on Jul. 22, 2008]. cited by applicant

E. Arthur Brown Company, "A Shooting Rest that Really Works .. ,"<http://www.eabco.comfTargetShooting01.html>, © 2007-2008, 1 pg. [Internet accessed Jul. 18, 2008]. cited by applicant

Really Right Stuff Vyce Photos, 3 Pages, At Least as Early As Jan. 1, 2018. cited by applicant

Vyce Equipment Support Mount, 2 Pages, At Least as Early As Jan. 1, 2018 (Internet Accessed May 25, 2018). cited by applicant

Hog Saddle PIG0311-G Field Tripod, 4 Pages, At Least as Early As Jan. 1, 2018 (Internet Accessed May 25, 2018). cited by applicant

MOD7 Hog Saddle, 4 Pages, At Least as Early As Jan. 1, 2018 (Internet Accessed May 25, 18). cited by applicant

Bog Super Steady Combo, 2 Pages, At Least as Early As Jan. 1, 2017 (Internet Accessed May 25, 2018). cited by applicant

Caldwell Shooting Supplies, Assembly and Usage Instructions for the Stable Table Shooting Bench, Instruction #1000196, Revision E, Jan. 1, 2017, 4 pages. cited by applicant

Caldwell Shooting Supplies, Assembly and Usage Instructions, BR Pivot Shooting Bench, Instruction #1001667, Jan. 1, 2017, 6 pages. cited by applicant

Herter's Deluxe Shooting Bench: Cabela's, 2 pages, accessed Jan. 8, 2018, <<https://www.cabelas.com/product/HERTERS-SHOOTING-BENCH/2201557.uts?slotId=0>>. cited by applicant

MSB200-Muddy Deluxe Shooting Bench, 4 pages, accessed Jan. 8, 2018. cited by applicant

Muddy, The Swivel Action Shooting Bench, 4 pages, accessed Jan. 8, 2018, <<https://shop.gomuddy.com/the-swivel-action-shooting-bench/>>. cited by applicant

Muddy, The Xtreme Shooting Bench, 4 pages, accessed Jan. 8, 2018. cited by applicant

Caldwell Shooting Supplies, DeadShot ChairPod, Usage and Assembly Instructions, Admitted Prior Art (at least as early as Jan. 2021), 8 pages. cited by applicant

Caldwell Shooting Supplies, DeadShot TreePod, Usage and Assembly Instructions, Admitted Prior Art (at least as early as Jan. 2021), 8 pages. cited by applicant

Battenfeld Technologies, Inc., Caldwell DeadShot ChairPod, YouTube video available at <https://www.youtube.com/watch?v=1HkgOjgGL8E>, Sep. 22, 2015, 1 page. cited by applicant

Optics Warehouse, Bog Pod Death Grip Tripod—Quickfire Review, YouTube video available at <https://www.youtube.com/watch?v=bbPc-RNG300>, Apr. 26, 2020, 1 page. cited by applicant

Nexcam Axis SG 360; <https://web.archive.org/we/20221007123252/https://nexcam.com/shop/nexcam-axi-sg-360-shooting-chair/> (Year: 2022). cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of and claims priority to U.S. application Ser. No. 17/659,222, filed Apr. 14, 2022, which claims priority to U.S. Provisional Application No. 63/175,361, filed Apr. 15, 2021, the entireties of which are hereby incorporated by reference.

FIELD

(1) The present disclosure generally relates to firearm shooting accessories, and more particularly to a stowable shooting rest chair that is adaptable for left and right handed shooters.

BACKGROUND

(2) Shooting rest chairs are used to support and stabilize a shooter firing a firearm and usually also to support and stabilize the firearm being fired. Some shooting rest chairs have limitations when it comes to being adaptable for left and right handed shooters.

SUMMARY

(3) In one aspect, a shooting rest chair for supporting a shooter and a weapon above a support surface comprises a base, a seat, a pivot hub, and a weapon support arm assembly. The base is configured to rest on the support surface. The seat is supported by the base, and the seat is sized and shaped to support the shooter when the shooter is sitting on the seat. The pivot hub is supported by the base and connects the base to the seat. The weapon support arm assembly is connected to the pivot hub by a first pivot connection. The weapon support arm assembly is configured to support the weapon. The weapon support arm assembly is pivotable with respect to the base and the seat about the first pivot connection.

(4) In another aspect, a portable chair for supporting a person above a support surface comprises a base and a seat. The base is configured to rest on the support surface and is configurable in a deployed configuration and in a collapsed configuration. The seat includes a holder under the seat, and the seat is supported by the base. The seat is sized and shaped to support the person when the person is sitting in the seat. The portable chair is configurable in an operational configuration and in a stowed configuration.

(5) Other objects and features of the present disclosure will be in part apparent and in part pointed out herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective of a shooting rest chair;
 - (2) FIG. 2 is a perspective of a base of the chair;
 - (3) FIG. 3 is a perspective of a seat and weapon support arm assembly of the chair;
 - (4) FIG. 4 is an exploded perspective of a connection of the base, seat, and weapon support arm assembly;
 - (5) FIG. 5 is an exploded perspective of a first pivot connector;
 - (6) FIG. 6 is a fragmentary cross-section of the connection of the base, seat, and weapon support arm assembly;
 - (7) FIG. 7 is a fragmentary cross-section of the base, seat, and weapon support arm assembly; and
 - (8) FIG. 8 is a bottom perspective of the shooting rest chair.
- (9) Corresponding reference numbers indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

- (10) Referring to FIG. 1, the shooting rest chair **10** is designed to support a sitting person (e.g., a

shooter) and a firearm to be fired by the person. As will become apparent, the shooting rest chair **10** is configurable between operational and stowed configurations. In the operational configuration, the shooting rest chair is adjustable to adapt for left-handed and right-handed shooters.

(11) Referring to FIGS. **1-3**, the shooting rest chair **10** generally includes a base **12** and a seat **14** (chair components) that together make up (broadly, “a chair”), and a weapon support arm assembly **16**. The seat and weapon support arm assembly are pivotable with respect to each other about a post **18** supported by the base which defines a generally vertical pivot axis PA. As will be explained in further detail, the post **18** pivotably supports the weapon support arm assembly **16** at the top of the base and below the seat **14** (broadly, “pivot connection”) to permit the weapon support arm assembly to rotate to left and right sides of the seat, thereby allowing the shooting rest chair **10** to be usable by left-handed and right-handed shooters.

(12) The base **12** includes a plurality of legs **22**, upper (first) and lower (second) post portions **18A**, **18B**, upper (first) and lower (second) collars **24A**, **24B**, and pivot links **26**. The upper post portion **18A** and upper collar **24A** form a pivot hub **20** to which the seat **14** and weapon support arm assembly are pivotably connected. Together, the upper and lower post portions **18A**, **18B** form the post **18** (column) that defines the pivot axis PA that permits pivotable rotation of the chair and the weapon support arm assembly. As will become apparent, the post portions are supported by the plurality of legs **22**. In the illustrated embodiment, the upper and lower post portions **18A**, **18B** can be extended (deployed) and retracted (collapsed) by a telescoping post connection. The upper collar **24A** is an unthreaded “O” collar that defines a hollow interior that extends around upper post portion **18A**. The upper collar **24A** includes an exterior having four pivot connectors, one for each leg, spaced apart around the exterior of the upper collar. Likewise, the lower collar **24B** is an unthreaded “O” collar that defines a hollow interior that extends around the lower post **18B**. The lower collar **24B** includes an exterior having four pivot connectors, one for each pivot link **26**, spaced apart around the exterior of the lower collar. The upper and lower collars **24A** and **24B** are sized and shaped to engage the respective upper and lower posts, **24A** and **24B**, respectively. In the illustrated embodiment, the upper collar **24A** is sized and shaped to be larger than the lower collar **24B**.

(13) Each of the legs **22** includes an upper (first) leg portion **22A** a lower (second) leg portion **22B**, and a leg connector collar **28**, and has an associated pivot link **26**. The upper leg portion **22A** includes a proximal end pivotably connected to the upper collar **24A**. The lower leg portion **22B** has a proximal end telescopically received in the distal end of the upper leg portion **22A**. The leg connector collar **28** assists the telescopic connection of the upper leg portion and the lower leg portion. Feet **30** are connected to the distal ends of the lower leg portions **22B** for contacting the ground (broadly, “support surface”). The pivot links **26** connect the leg connector collar **28** to the pivot connectors of the lower collar **24B**. Retainers **32** (e.g. spring biased detents) carried by the lower leg portions **22B** are receivable in a series of openings in the upper leg portions **22A** for selectively adjusting a length of each leg.

(14) The base **12** is configurable between a deployed configuration and a collapsed configuration. FIGS. **1, 2**, and **8** show the base **12** in the deployed configuration. In the deployed configuration, the legs are arranged to engage the support surface to support the seat above the support surface. For example, the support surface can be an exterior surface such as turf, pavement, and/or gravel, etc. or can be an interior surface such as a floor.

(15) When the base **12** is in the deployed configuration, the four legs **22** are extended outward from the pivot hub **20** to the ground. Each leg forms about a 45-degree angle with respect to the pivot axis PA. The pivot links **26** are extended in a direction transverse the pivot axis PA to provide support to the base via connection to the lower post portion **18B**. It will be appreciated that the upper post **18A** protrudes upward to be exposed above the upper collar **24A**. Bases having other configurations (e.g., more or fewer legs) can be used without departing from the scope of the present disclosure.

(16) When the base **12** is in a collapsed configuration (not shown), the four legs are folded inward toward the lower post portion **18B** so each leg is about parallel with respect to the pivot axis PA. The lower post portion **18B** is retracted into the upper post portion **18A** and the pivot links **26** are pivoted

inward toward the pivot axis PA. Further, the lower leg portion 22B is retracted into the upper leg portion 22A such that the detent carried by the proximal end of the lower leg portion is generally near the proximal end of the upper leg portion. As will be explained below, when in the collapsed configuration, the base 12 is sized and shaped to fit in a holder 34 located under the seat 14. FIG. 8 shows two instances of the holder 34, one connected to the seat, and another removed from the seat. (17) Referring to FIGS. 1, 3, and 4, the weapon support arm assembly 16 is configured to rotate about the upper post portion 18A (pivot axis PA) to left and right sides of the seat 14 for respective left and right handed shooters. The weapon support arm assembly 16 comprises an arm 36 and a weapon holder (broadly, “weapon rest”) 38. The arm comprises first, second, and third arm portions, 40, 42, and 44. The first arm portion includes a first segment 40A and a second segment 40B. A proximal end of the first segment 40A includes a first pivot connector 40C configured to pivotably connect to the upper post portion 18A when the upper post portion is in the deployed configuration. The first segment 40A extends distally from the first pivot connector 40C located at the pivot hub 20 laterally outward a distance sufficient to extend outboard of a periphery of the seat. The first segment 40A has a length longer than a length of the center of the seat 14 to an edge of the seat. The second segment 40B of the first arm portion 40 includes a proximal end connected to the first arm segment 40A below the seat 14, and the second segment 40B extends distally to a distal end above the seat. The second segment 40B is generally upstanding (e.g., generally parallel with the generally vertical pivot axis PA). Other configurations of a weapon support arm assembly can be used without departing from the scope of the present disclosure.

(18) Referring to FIGS. 5-7, the pivot connector 40C includes a bushing 46 and an adjustable collar 48. The adjustable collar 48 may be welded to the proximal end of the first segment 40A. The bushing 46 is sized and shaped to be received in the collar 48 and engage an inner surface 48A of the collar. The bushing 46 may be made from Delrin or a POM material that snaps into the collar to prevent movement. The collar 48 (e.g., made of metal) includes a slot, a cam lever 41, a fastener 43 (e.g., bolt), and a nut 45 (broadly, “adjusting components.”) The cam lever is arranged to reduce a width of the slot when in a closed position (e.g., FIG. 5) and to permit the slot width to increase when in an open position. The nut 45 is threaded onto the fastener 43 such that the cam lever 41 cams against the collar 48 to draw flanges of the collar toward each other to reduce the width of the slot between the flanges. After initially adjusting the nut 45 with a wrench, the cam lever 41 can be used to permit tightening or loosening of the collar 48 without tools. The collar 48 may be tight to prevent pivoting of the weapon support with respect to the seat, or can be loose to permit pivoting of the weapon support with respect to the seat, per the preference of the user. The location of the first pivot connector 40C (at the pivot hub 20) permits the first segment 40A, and thus the weapon support arm assembly 16, to rotate freely about the pivot axis PA.

(19) Referring to FIGS. 1 and 3, the second arm portion 42 is arranged between the first and third arm portions 40, 44. (Only a small portion of the second arm portion 42 is depicted in the FIGS. 1 and 3.) The second arm portion 42 permits the shooter to selectively raise or lower the height of the weapon holder 38. The second arm portion 42 includes a tube having a lower end, an upper end, and a length extending therebetween. The second arm portion 42 is telescopically received in the second segment 40B such that the lower end of the tube remains within the second segment 40B. A retainer 50 (e.g. rotatable friction knob) is connected to the second segment 40B and permits the user to selectively lock the second arm portion 42 to set a distance the second arm portion extends upward out of the second segment 40B. Other ways of adjusting the height of the weapon holder (e.g. a detent, etc.) can be used without departing from the scope of this disclosure.

(20) The third arm portion 44 includes first and second sections 44A, 44B each having respective opposite ends. The first section 44A includes a second pivot connector 44C for connecting the first section 44A to the second arm portion 42. The first section 44A includes a third pivot connector 44D for connecting to the second section 44B. The second section 44B includes a fourth pivot connector 44E for connecting to the weapon holder 38. In the illustrated embodiment, the first and second sections 44A, 44B via the second and third pivot connectors 44C, 44D permit the shooter to

selectively adjust the lateral position of the weapon holder **38** with respect to the seat **14**.

(21) The weapon holder **38** includes a cradle comprising a bed **51** and opposing jaws **53** for clamping a weapon (e.g. firearm). The weapon holder **38** includes a clamp actuator **55** (e.g., rotatable knob) configured to move the first and second jaws **53** toward and away from each other to clamp and release the weapon. The weapon holder **38** is adjustable (e.g. pivotable about multiple axes) to change an orientation of the weapon with respect to the chair **10**. Other types of weapon holders (e.g. without clamping) can be used without departing from the scope of the present disclosure. As will be discussed below, the weapon support arm assembly **16** (e.g., including the weapon holder **38**, and arm portions **40**, **42**, **44**) and the seat **14** are independently pivotable with respect to each other such that both can rotate 360 degrees (broadly, at least 180 degrees, or at least 90 degrees) with respect to each other. The weapon support arm assembly **16** can pivot in front of the seat **14** to left or right sides of the seat and can pivot behind the seat.

(22) Referring to FIGS. **3** and **4**, the seat **14** includes a seat frame **52** and a seat panel **52A** connected thereto. In the illustrated embodiment, a backrest including a backrest frame **54** and a backrest panel **54A** is pivotably connected to the seat **14**. The backrest is shown in FIG. **3** in a deployed configuration in which it extends upward from the seat **14**. It will be appreciated that the backrest is foldable downward toward the seat **14** to a folded configuration. Support straps connect the seat frame **52** to the backrest frame **54** to support the backrest in the deployed configuration. The seat frame **52** includes a seat tube **56** (broadly, “pivot hub connector”) for connecting the seat frame **52** to the upper post portion **18A**. The seat tube **56** is a hollow tube welded to the frame and sized and shaped for sliding over the upper post portion **18A**. As shown in FIG. **6**, when the seat tube **56** is received over the upper post portion, the arm connector **40C** (including the collar **48** and bushing **46**) are received over the seat tube **56**, such that both the seat and support arm assembly **16** are pivotably connected to the pivot hub **20**. The seat **14**, support arm assembly **16**, and base **12** can be disassembled by releasing the cam lever **41**, and separating the base downward from the seat and support arm assembly, and separating the support arm assembly downward from the seat. A spring detent **61** including a lug **63** (FIG. **4**) and a spring **65** (FIG. **6**) is provided to prevent inadvertent upward removal of the seat **14** from the base **12**, and the lug can be pressed inward by the user to permit removal of the seat tube **56** past the lug.

(23) Referring to FIGS. **7** and **8**, the holder **34** is connected to the seat frame **52** for holding the base **12** when the base is in a collapsed configuration. In the illustrated embodiment, the holder **34** includes a container formed of a flexible fabric forming a bag-like enclosure. A zipper **67** may be provided for opening and closing the holder **34**. The holder may include grommets **69** (openings) through which fasteners are passed to secure the holder to the frame **52**. In other embodiments, the holder may include straps (e.g. Velcro®, button snaps, etc.) fastened to the frame **52** for holding the base **12** to the seat when in the stowed configuration.

(24) It will be understood that the shooting rest chair **10** is configurable to operational and stowed configurations. To configure the shooting rest chair **10** in the operational configuration, the base is configured in its deployed configuration as described above. With the cam lever **41** in its open position, the user places the weapon support arm assembly **16** over the seat tube **56** such that an outer surface of the seat tube engages an inner surface of the bushing **46**. The user then places the seat tube **56** over the upper post portion **18A** such that the seat and support arm assembly **16** are pivotably connected to the pivot hub **20**. As the seat tube **56** is installed over the upper post portion **18A**, the user presses the detent **61** (retainer) inward and it resiliently extends outward when the seat tube has passed over the detent. The detent **61** serves to retain the seat **14** on the upper post portion **18A** when in the operational configuration. The collar **48** of the weapon support arm assembly **16** is secured using the cam lever **41**, such that the collar is permitted to pivot relative to the seat tube or locked relative to the seat tube, depending on the setting of the nut **45** on the fastener **43**. To disassemble the shooting rest chair **10** and configure it in the stowed configuration, the user first unlocks the cam latch. The user then presses the detent with the user's finger and lifts the seat **14** up from the upper post portion **18A**. Next, the user removes the weapon support arm assembly **16** by lifting it up from

the upper post portion **18A**. The user then collapses the base **12** to the collapsed configuration as described above and places the collapsed base into the holder **34** under the seat **14**.

(25) It will be apparent that modifications and variations are possible without departing from the scope of the invention defined in the appended claims. The dimensions and proportions described herein are by way of example without limitation. Other dimensions and proportions can be used without departing from the scope of the present disclosure.

(26) As various changes could be made in the above constructions and methods without departing from the scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A shooting rest chair for supporting a shooter and a weapon above a support surface, the shooting rest chair comprising: a base configured to rest on the support surface, the base being configurable in a deployed configuration and in a collapsed configuration; a seat supported by the base, the seat being sized and shaped to support the shooter when the shooter is sitting on the seat; and a weapon support arm assembly supported by the base, the weapon support arm assembly being configured to support the weapon, the weapon support arm assembly being pivotable with respect to the seat through a range of motion, the weapon support arm assembly being configured to be selectively fixable at least three different positions along the range of motion to prevent the weapon support arm assembly from pivoting with respect to the seat.
2. The shooting rest chair of claim 1, further comprising a retainer configured to selectively fix the position of the weapon support arm assembly along the range of motion to prevent the weapon support arm assembly from pivoting with respect to the seat.
3. The shooting rest chair of claim 2, wherein the retainer is part of the weapon support arm assembly.
4. The shooting rest chair of claim 2, wherein the retainer comprises a clamp.
5. The shooting rest chair of claim 2, wherein the retainer comprises a cam lever moveable between a locked position where the weapon support arm assembly is held in a fixed position along the range of motion and an unlocked position where the weapon support arm assembly is free to pivot along the range of motion.
6. The shooting rest chair of claim 5, wherein the retainer comprises a collar, the collar being operatively connected to the cam lever such that the collar is tightened when the cam lever is in the locked position and loosened with the cam lever is in the unlocked position.
7. The shooting rest chair of claim 6, further comprising a pivot hub supported by the base, the collar positioned on the pivot hub and configured to rotate about the pivot hub when the cam lever is in the unlocked position and to be fixed in position on the pivot hub when the cam lever is in the locked position.
8. The shooting rest chair of claim 2, wherein the retainer forms a pivot connection about which the weapon support arm assembly pivots along the range of motion.
9. The shooting rest chair of claim 2, wherein the range of motion permits pivoting of the weapon support arm assembly between a left side of the seat and a right side of the seat to change a configuration of the weapon support arm assembly between a left handed shooter arrangement and a right handed shooter arrangement.
10. The shooting rest chair of claim 9, wherein the range of motion extends across a front of the seat to permit pivoting of the weapon support arm assembly between the left side of the seat and the right side of the seat across a front of the seat to change the configuration of the weapon support arm assembly between the left handed shooter arrangement and the right handed shooter arrangement.
11. The shooting rest chair of claim 2, wherein the range of motion about which the weapon support arm assembly is pivotable with respect to the seat comprises at least 90 degrees.
12. The shooting rest chair of claim 2, wherein the range of motion about which the weapon support arm assembly is pivotable with respect to the seat comprises at least 180 degrees.

13. The shooting rest chair of claim 2, wherein the range of motion about which the weapon support arm assembly is pivotable with respect to the seat comprises 360 degrees.
14. The shooting rest chair of claim 2, wherein the weapon support arm assembly is configured to be selectively fixable at any position along the range of motion to prevent the weapon support arm assembly from pivoting with respect to the seat.
15. The shooting rest chair of claim 2, wherein the seat is pivotable with respect to the base, and the weapon support arm assembly is pivotable with respect to the base independent from the seat.
16. The shooting rest chair of claim 2, wherein the base comprises a plurality of legs configured to engage the support surface when the base is in the deployed configuration.
17. The shooting rest chair of claim 2, wherein the weapon support arm assembly includes an arm and a weapon rest, and wherein the arm is configured to extend under the seat and upward to the weapon rest.
18. The shooting rest chair of claim 17, wherein the arm includes a first arm portion, a second arm portion, and a third arm portion, the first arm portion being coupled to the second arm portion by a height adjustment connection and the second arm portion being connected to the third arm portion by a second pivot connection.
19. The shooting rest chair of claim 17, wherein the weapon rest is connected to the arm by a pivot connection.
20. The shooting rest chair of claim 19, wherein the rest comprises a cradle for holding the weapon.
21. The shooting rest chair of claim 1, wherein said at least three different positions along the range of motion are non-predetermined positions along the range of motion.
22. The shooting rest chair of claim 1, wherein in the deployed position the base is arranged to support the shooting rest chair on the support surface and in the collapsed configuration the base is arranged to be stowed.
23. A shooting rest chair for supporting a shooter and a weapon above a support surface, the shooting rest chair comprising: a base configured to rest on the support surface; a seat supported by the base, the seat being sized and shaped to support the shooter when the shooter is sitting on the seat; and a pivot hub supported by the base; a weapon support arm assembly supported by the base, the weapon support arm assembly being configured to support the weapon, the weapon support arm assembly being pivotable with respect to the seat about the pivot hub through a range of motion, the weapon support arm assembly being configured to be selectively fixable at at least three different positions along the range of motion to prevent the weapon support arm assembly from pivoting with respect to the seat; a retainer configured to selectively fix the position of the weapon support arm assembly along the range of motion to prevent the weapon support arm assembly from pivoting with respect to the seat, the retainer including a first portion and a second portion, the first and second portions configured to move relative to the pivot hub to fix the position of the weapon support arm assembly along the range of motion to prevent the weapon support arm assembly from pivoting with respect to the seat.
24. The shooting rest chair of claim 23, wherein the first and second portions of the retainer are configured to engage the pivot hub in an arrangement where the first portion and second portion engage opposing sides of the pivot hub to fix the position of the weapon support arm assembly along the range of motion.
25. The shooting rest chair of claim 24, wherein the pivot hub is a pivot hub connector of the seat, the pivot hub connector pivotably coupling the seat to the base to permit the seat to pivot relative to the base.
26. The shooting rest chair of claim 24, wherein the first and second portions of the retainer are configured to be drawn toward the pivot hub to constrict the pivot hub between the first and second portions to fix the position of the weapon support arm assembly along the range of motion.
27. The shooting rest chair of claim 23, wherein the first and second portions of the retainer bound an opening sized and shaped to receive the seat, the first and second portions being configured to move relative to each other to decrease a size of the opening to fix the position of the weapon support arm

assembly along the range of motion and to increase the size of the opening to permit the weapon support arm assembly to move along the range of motion.

28. The shooting rest chair of claim 23, wherein the retainer includes an actuator configured to configure the first and second portions of the retainer relative to each other to fix the position of the weapon support arm assembly along the range of motion.

29. The shooting rest chair of claim 28, wherein the actuator is moveable between a locked position where the weapon support arm assembly is held in a fixed position along the range of motion by the first and second portions of the retainer and an unlocked position where the weapon support arm assembly is free to pivot along the range of motion.

30. The shooting rest chair of claim 29, wherein the actuator comprises a cam lever.

31. The shooting rest chair of claim 28, wherein the retainer comprises a collar, the actuator being configured to tighten the collar to configure the first and second portions to fix the position of the weapon support arm assembly along the range of motion and to loosen the collar to permit the weapon support arm assembly to move along the range of motion.

32. The shooting rest chair of claim 31, wherein the actuator comprises a cam lever.

33. The shooting rest chair of claim 31, wherein the collar is supported by the pivot hub and configured to rotate about the pivot hub when the collar is loosened and to be fixed in position with respect to the pivot hub when the collar is tightened to fix the position of the weapon support arm assembly.

34. The shooting rest chair of claim 31, wherein the first and second portions are coupled to and inhibited from moving relative to the collar.

35. The shooting rest chair of claim 31, wherein the first and second portions are snapped into the collar.

36. The shooting rest chair of claim 23, wherein the retainer comprises a collar, the collar being configurable to tighten the collar to configure the first and second portions to fix the position of the weapon support arm assembly along the range of motion, and the collar being configurable to loosen the collar to permit the weapon support arm assembly to move along the range of motion.
